

Question	Answer	Marks
5(a)	$E = F/Q$	M1
	$F = ma$ and (so) $q/m = a/E$	A1
5(b)	$m = (4 \times 1.60 \times 10^{-19} \times 3.5 \times 10^4) / 1.5 \times 10^{12} (= 1.49 \times 10^{-26} \text{ kg})$	B1
	$= 1.49 \times 10^{-26} / 1.66 \times 10^{-27} = 9.0 \text{ (u)}$	A1
5(c)	protons: 4 and neutrons: 5	A1
5(d)(i)	nuclei have the same charge and so same (magnitudes of) force	B1
5(d)(ii)	nuclei have different masses and same force and so different (magnitudes of) acceleration	B1

Question	Answer	Marks
6(a)	(coulomb is an) ampere second	B1